

On the Switching Parameter Variation of Metal Oxide RRAM—Part II: Model Corroboration and Device Design Strategy

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Abstract—Using the model developed in Part I of this two-part paper, the simulated dc sweep and pulse transient characteristics of a metal oxide resistive random access memory cell are corroborated with the experimental data of HfO_x memory. Key switching features such as the abrupt SET process, gradual RESET process, current fluctuation in the RESET process, and multilevel resistance state distributions are captured by the simulation. The current fluctuation in the RESET process is caused by the competition between the simultaneous oxygen vacancy recombination and generation processes. The origin of the high-resistance state variation and the tail bit problem are attributed to the variation of the tunneling gap distances and the stochastic nature of new Vo generation in the tunneling gap region, respectively. The use of the write-verify technique and a bilayer oxide structure are proposed to achieve a tighter resistance distribution.

Index Terms—Parameter fluctuation, resistive random access memory (RRAM), resistive switching, switching uniformity, tail bit, variability.

I. INTRODUCTION

THE METAL oxide resistive random access memory (RRAM) is extensively studied as a competitive candidate for future nonvolatile memory applications due to its simple structure, fast switching speed, great scalability, and compatibility with silicon complementary metal-oxide-semiconductor technology [1]–[3]. Among dozens of switching oxide materials, NiO - [4], WO_x - [5], HfO_x - [6], [7], TaO_x - [8], [9], and AlO_x [10]-based memory devices have shown excellent reliability, including long endurance and stable retention at elevated temperature. A 4-Mb macro chip based on HfO_x memory has been demonstrated [11]. However, the key obstacle for metal oxide RRAM to becoming mature for large-scale manufacturing is the relatively poor control of switching uniformity.

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Significant parameter fluctuations exist in the switching voltages and the resistance distributions, which include temporal fluctuations (cycle to cycle) and spatial fluctuations (device to device). A large memory array usually contains noticeable tail bits in the resistance distribution [12], which remarkably reduces the resistance window and therefore imposes a great challenge to realizing the multilevel capability of the metal oxide RRAM technology.

The origin of the variations in the resistive switching is not well understood yet, despite the fact that the variations are observed in all RRAM reported. Phenomenological models such as the network of random circuit breakers [13] have been applied to the unipolar RRAM devices. In this paper, this problem is studied for the metal-oxide-based bipolar RRAM devices. Here, unipolar means the set from the high-resistance state (HRS) to the low-resistance state (LRS) and the reset from LRS to HRS occur at the same voltage polarity, and bipolar means the set and the reset occur at opposite voltage polarities. Through the development of a physics-based stochastic simulator to quantify the switching variations, we identify the major sources of variations and fluctuations. Specifically in this paper, we focus on the cycle-to-cycle variability in RRAM devices, which is a more intrinsic problem than the device-to-device variability, because the device-to-device variability depends on the fabrication processes, and better uniformity control of the processes may improve the device-to-device uniformity. For example, a chemical-mechanical polish was adopted to smooth the bottom electrode and improve the uniformity HfO_x -based RRAM devices in an array [6]. The obtained insights provide guidelines for future device design and optimization against the undesired switching parameter variations. In part I of the paper [14], we describe the developed model, which includes the trap-assisted-tunneling current solver and the stochastic generation and recombination of oxygen vacancies (Vo).¹ The considerations and approximations to reproduce the switching characteristics of a bipolar RRAM device are discussed. In part II of the paper (the current paper), we corroborate the simulation results with the experimental data of HfO_x -based RRAM devices [15], [16], such as the dc I - V switching characteristics and pulse transient switching waveform, multilevel resistance distribution, etc., with a focus on discussing the origin of the tail bit problem and possible elimination methods such as the use of the write-verify technique and a bilayer oxide device structure.

¹In the subsequent discussions in this paper, Vo can represent a single oxygen vacancy or multiple oxygen vacancies.

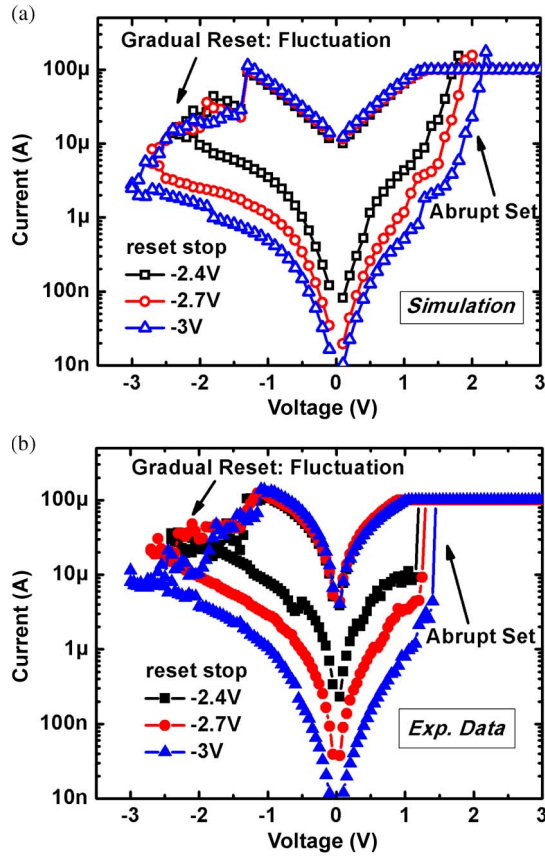


Fig. 1. (a) Simulated and (b) experimental I - V characteristics of HfO_x memory for different reset stop voltages. Abrupt SET and gradual RESET processes are reproduced by the simulation.

II. EXPERIMENTS

$\text{TiN}/\text{HfO}_x/\text{Pt}$ bipolar RRAM devices were fabricated. Pt of 50 nm was first deposited by electron-beam evaporation on a silicon substrate as the bottom electrode layer. Then, 10 nm HfO_x was deposited by atomic layer deposition using tetrakis (ethylmethylamino) hafnium (TEMA-Hf $\text{Hf}[\text{N}(\text{C}_2\text{H}_5)(\text{CH}_3)]_4$) and H_2O as precursors at 220 °C. The crossbar patterns with a $0.5 \times 0.5 \mu\text{m}^2$ active cell area were defined by photolithography. Then, 50-nm TiN was deposited by reactive sputtering and was lifted off as a top electrode layer. The electrical measurements were performed with an Agilent 4156C semiconductor parameter analyzer, an Agilent 81101A pulse generator, and a Tektronix DPO 4054 oscilloscope. The bottom electrode (Pt) was grounded, and the signals were applied to the top electrode (TiN) in all the measurements.

III. SIMULATION RESULTS AND CORROBORATION WITH EXPERIMENTS

In Part I of the paper [14], we described a stochastic simulator that can produce the dc switching characteristics of a bipolar RRAM device. Here, we present additional dc sweep simulation results, as plotted in Fig. 1(a), which shows the effect of different reset stop voltages on the HRS resistance. It is shown that larger reset stop voltages lead to a higher HRS resistance because more oxygen ions are driven back to the

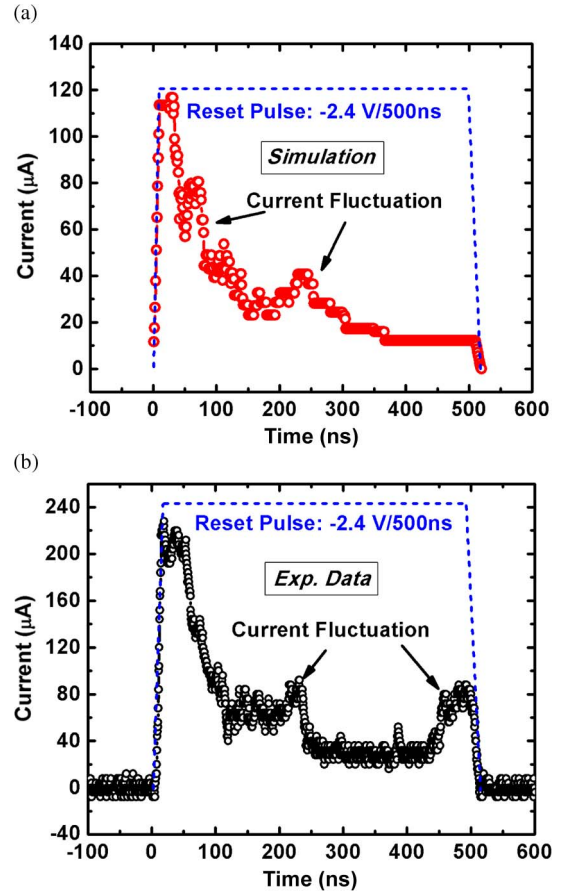


Fig. 2. (a) Simulated and (b) experimental pulse transient current in the RESET process. Current fluctuation is reproduced by the simulation. In (b), the last current jump at the end of the pulse is deduced to be a new generation of V_o in the gap region.

bulk oxide and result in a larger ruptured conductive filament (CF) length, and the current in HRS is mainly modulated by the tunneling gap distance, as explained in Part I of the paper [14]. The corresponding experimental data of our fabricated HfO_x memory are shown in Fig. 2(b). Experimentally, modulation of the HRS resistance by controlling the reset stop voltages is also reported by others for HfO_x and other metal oxide materials [17], [18], suggesting that it is a common property of metal oxide RRAM devices that can enable the multilevel storage capability. In addition to the HRS modulation property, it is noticed that our dc sweep simulation result also captures other detailed features present in an experimental measurement such as a more abrupt transition in the SET process and a more gradual transition in the RESET process. The abrupt SET process is due to the positive feedback that increasing current leads to raised temperature, thus larger V_o generation probability and, in turn, higher current. The gradual RESET process is due to the gradual dissolution of the CFs away from the electrode. We also perform the simulation under a pulse programming condition, and the RESET transient waveform is reproduced, as plotted in Fig. 2(a). The corresponding experimental RESET transient waveform is shown in Fig. 2(b). It is shown that a significant current fluctuation is present in the RESET pulse transient waveform (it is also observable in a dc RESET curve shown above). We track the V_o evolution

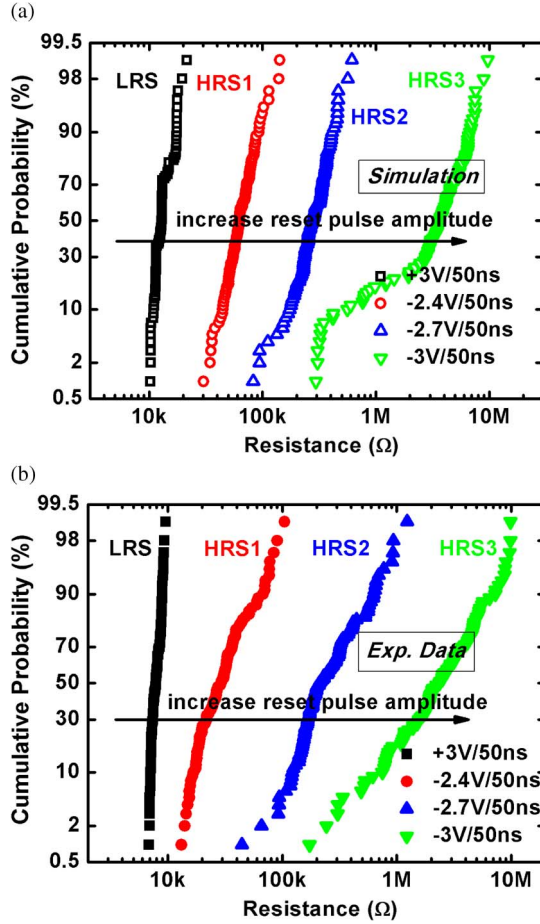


Fig. 3. (a) Simulated and (b) experimental multilevel states achieved by linearly increasing RESET pulse amplitude.

with the current evolution simultaneously and find that the current jump corresponds to new V_o generation in the gap region. Therefore, the current fluctuation during the RESET process is caused by a competition between the simultaneous V_o generation and recombination processes. Although the CFs are partially ruptured, the high field present in the gap region may also induce V_o generation. This is particularly prominent in the beginning stage of the current drop when the gap size is small and the field and temperature are both high. The generated V_o in the gap tend to reconnect the ruptured filament with the cathode, giving rise to a current jump. We have to point out that the observed current fluctuation is not due to a random telegraph-like noise current that is caused by the individual hopping electron tunneling, because the noise current under a small constant voltage on our sample is measured to be of less than 10% fluctuation [19]. Therefore, the larger fluctuation shown in the RESET transient waveform is indeed caused by the change of V_o configuration. We further perform the simulation for different pulse programming schemes for achieving multilevel HRS. In one scheme, the pulsewidth is fixed, and the pulse amplitude is linearly increased to achieve different HRS resistances, as shown in Fig. 3(a) [the corresponding experimental data are shown in Fig. 3(b)]. In an alternate scheme, the pulse amplitude is fixed, and the pulsewidth is exponentially increased, as shown in Fig. 4(a) [the corresponding experimen-

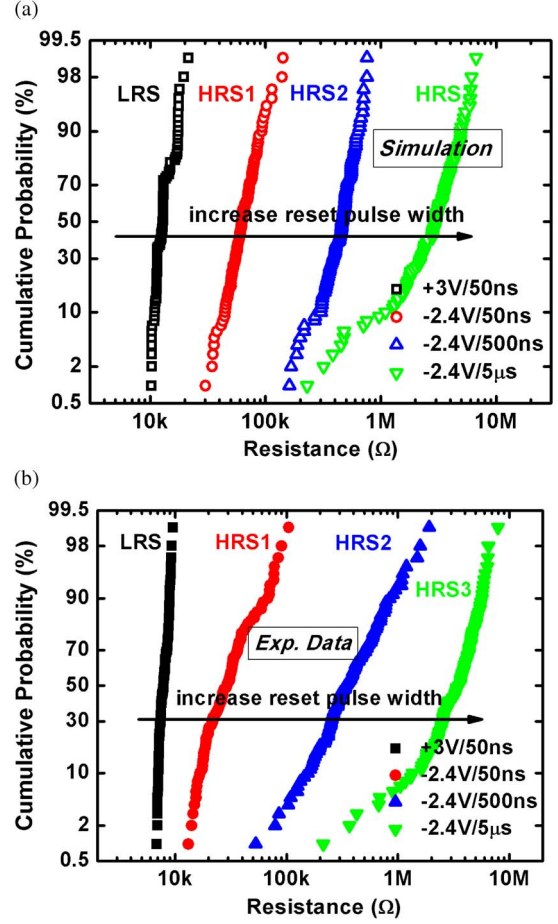


Fig. 4. (a) Simulated and (b) experimental multilevel states achieved by exponentially increasing RESET pulsewidth.

tal data are shown in Fig. 4(b)]. It has been suggested [15] that these two programming schemes are equivalent to achieve similar multilevel HRS owing to the exponential voltage–time relationship: the switching time exponentially decreases with the increase in applied voltage [20]. Our simulation further confirms that similar gap distances are achieved by the two programming schemes for a particular HRS level: for the three HRS levels shown in Figs. 5 and 7, the average gap distances are 0.7, 1.5, and 2.2 nm, respectively.

IV. DISCUSSION ON THE SWITCHING UNIFORMITY IMPROVEMENT STRATEGIES

Significant switching parameter variations have become the major problem that hinders metal oxide RRAM from large-scale manufacturing. In the literature, efforts have been reported to improve the uniformity either through materials/device structure engineering, e.g., to stabilize the formation/rupture of CFs by the local electric field enhancement effect through dopant diffusion [21], ion implantation [22], or inserting nanocrystals [23] into the bulk oxide. The good news is that with the down-scaling of the cell size, the switching uniformity, particularly the LRS resistance distribution, is noticeably improved due to the direct confinement of the conduction paths, as illustrated in the novel device structure [24] for a 50-nm NiO RRAM

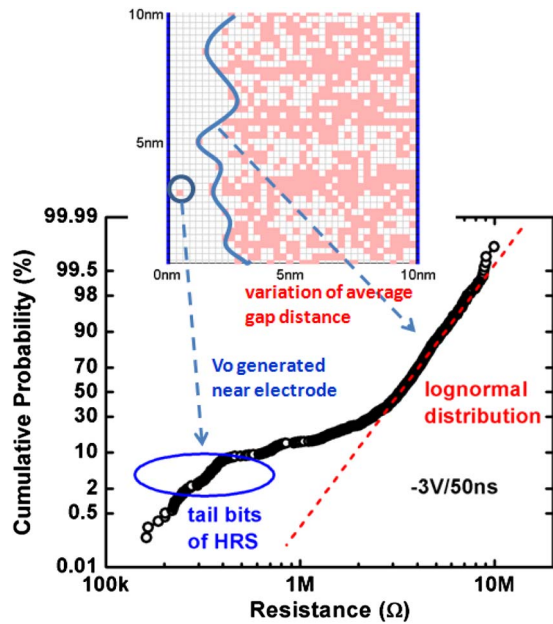


Fig. 5. Simulated HRS distribution after 1000 times pulse cycling. The lognormal distribution is due to the Gaussian distribution of the average gap distances. The tail bits are due to new V_o near the electrode generated at the end of the pulse (see Fig. 2(b) for an experimental example of tail bit creation).

device. However, the HRS resistance distribution is still widely spread even for this device [24] with a confined conduction path. Even for a 10-nm HfO_x RRAM device [25], the HRS variation is still remarkable. The reason may be attributed to the different origins of LRS and HRS resistance variations: For LRS, the current level is approximately linearly proportional to the number or the size of CFs; thus, the reduction of possible filament paths effectively confines the active switching area and thus reduces the variation. For HRS, the current level is almost exponentially dependent on the ruptured CF length; thus, any small variation of the tunneling gap distance may be magnified to be a remarkable variation of the HRS resistance, and the tunneling gap distance is not directly related to the area of the filament paths. Therefore, in this paper, we mainly focus on the HRS variation, which is expected to be an intrinsic problem even for future ultrascaled RRAM devices. We perform the pulse cycling simulation in a single device for 1000 times and plot the HRS distribution in Fig. 5. A straight line in this plot means a lognormal distribution of the HRS bits, which reflects a Gaussian-like variation of the average tunneling gap distances. At the same time, a noticeable tail of the distribution is shown in the plot. We correlate these tail bits with their corresponding V_o configuration and reveal that the tail bits correspond to the existence of V_o in the gap region. As analyzed above, during the RESET process, the recombination and generation processes of V_o compete with each other; thus, there is a possibility that a new V_o is created in the gap region, leading to the current jump observed in the transient current waveform. If the V_o generation occurs in the middle of a programming pulse, it has the chance to be recombined again with the mobile oxygen ions. However, if the V_o generation occurs just at the end of a programming pulse, it has no chance to be recombined again since the pulse is withdrawn afterward [see the experimentally

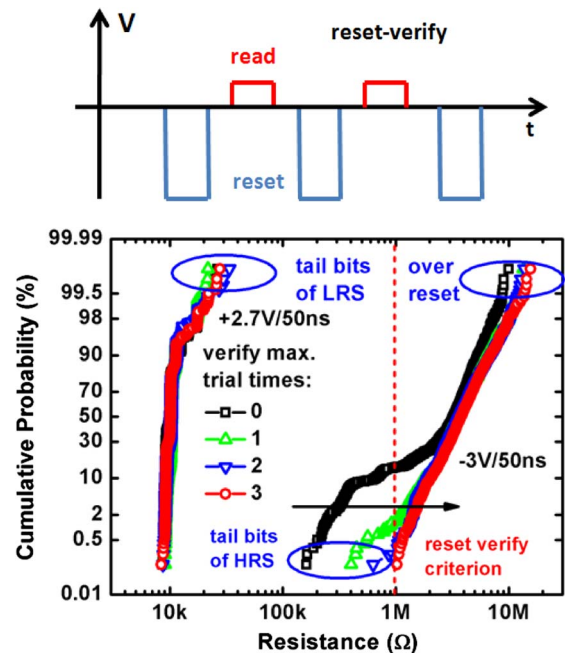


Fig. 6. Simulated LRS and HRS distributions with the write-verify technique. The purpose of multiple RESET pulses is to decrease the probability to get a new V_o generation at the end of a single pulse. With a maximum of three trials, above 99.9% HRS meet the criterion (1 MΩ). However, the over-RESET bits may in turn cause more tail bits of LRS, which is undesired.

observed last current jump in Fig. 2(b)]. Therefore, V_o are left in the gap region and act as a bridge for the electron tunnel from the electrode to the residual of the CFs. To remove the tail bits in HRS, the write-verify technique (write the cell first and then read; if the cell resistance does not meet the resistance criterion, write it again [26]) is usually used. The effectiveness of the write-verify technique is illustrated by the simulation results shown in Fig. 6. A maximum of three trials is able to make 99.9% HRS bits meet the criterion. However, it is noticed that the write-verify technique would introduce over-RESET bits in HRS, which, in turn, may cause more tail bits (with higher resistance than targeted) in LRS since the over-RESET makes it harder for the CFs to reconnect with two electrodes again. Of course, we can also use the write-verify technique for the SET process to reduce the tail bits in LRS. However, the cost of write-verify is very expensive because it decreases the writing speed and increases the energy consumption. Therefore, it is desirable to solve the problem through a combination of device structure innovation and programming technique. We propose using a bilayer oxide structure with a buffer oxide layer that has a larger oxygen ion migration barrier than the active switching oxide layer (see Fig. 7). The idea is to confine the location of resistive switching by completely rupturing the CFs in the active switching layer since the oxygen ion migration would stop at the interface of the two layers due to a larger migration barrier in the buffer layer. By tuning the active oxide layer thickness, the over-RESET bits are eliminated, which, in turn, reduce the tail bits in LRS. The results of simulating such a structure are shown in Fig. 7. However, the bilayer structure does not help reduce the tail bits in HRS because the new V_o generation problem is still present, and actually, it is more serious due to the reduced resistance window for

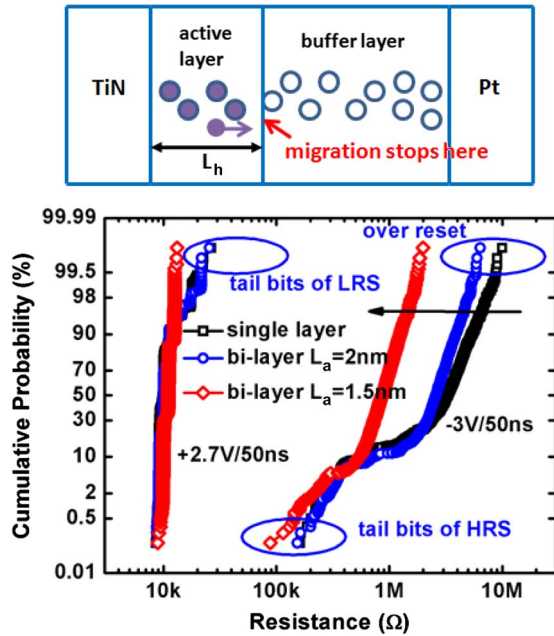


Fig. 7. Simulated LRS and HRS distributions of a bilayer structure. The buffer layer has a larger oxygen migration barrier (1.2 eV) than the active layer (1 eV) to confine the CF rupture. The total thickness remains the same (10 nm). It is shown that a 1.5-nm active layer is effective for reducing the over-RESET bits and the tail bits of LRS, but it does not help eliminate the tail bits of HRS.

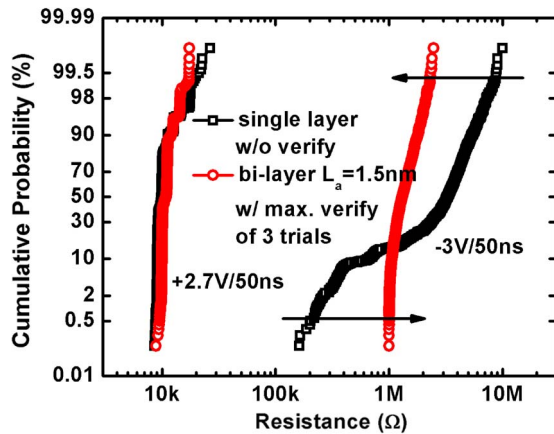


Fig. 8. Simulated both uniform LRS and HRS distributions are achieved by combining a bilayer structure with the write-verify technique.

the thinner active layer structure. Therefore, by combining the bilayer structure with the write-verify technique, it is expected to achieve both uniform LRS and HRS distributions, as simulated in Fig. 8. Experimentally, uniformity improvement has been demonstrated in the bilayer structure, such as $\text{HfO}_x/\text{AlO}_x$ [27], [28] and $\text{HfO}_x/\text{ZrO}_x$ [29]. The choice of the buffer layer materials and the optimization of thickness of each layer need further study in the future.

V. CONCLUSION

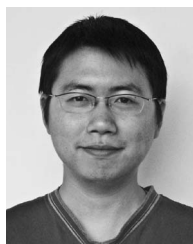
The dc sweep and pulse transient simulation results given by the stochastic simulator developed in Part I of the paper [14] are corroborated with the experimental data of HfO_x memory. Key switching features such as the abrupt SET process, gradual

RESET process, and current fluctuation in the RESET process are captured by the simulation. The experimental multilevel states achieved by different programming schemes are also reproduced by the simulation. The origin of the HRS resistance variations and the tail bit problem are attributed to the variations of the tunneling gap distances and new Vo generation in the ruptured CF region. The write-verify technique and a bilayer oxide structure are proposed to achieve a tight distribution in HRS and LRS resistances. This paper provides new understanding of the physical sources of the switching parameter variation phenomenon in metal oxide RRAM and guidelines for future device optimization. Furthermore, the developed simulator set up a platform to study other properties of RRAM devices such as retention [30], overshoot, etc., which will be our future work.

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